

The φ -Block of the AR(2) Sampler Targets the Wrong Full Conditional: Exact versus Conditional Likelihood in `updatePhiAR2TS`

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Abstract

The Metropolis–Hastings update for $\varphi = (\varphi_1, \varphi_2)$ in `update_params_ar2.R` evaluates the Box–Jenkins *conditional* likelihood, while the latent-block updaters evaluate the *exact* stationary likelihood. The two Gibbs blocks therefore target different distributions; this note proves the discrepancy, explains why the analogous omission is exactly harmless in the original AR(1) code, quantifies the effect at $n_t = 50$, and states the three-line fix.

1 Setup and notation

- Fix one latent channel (z^* , τ^* , or a knot coordinate) and one knot k ; write $Y_{1:T}$ for its copula-transformed series, with independent replicates across $k = 1, \dots, K$.
- The AR(2) prior on $Y_{1:T}$ is the stationary causal Gaussian AR(2) normalised to unit marginal variance, which by Yule–Walker pins

$$\gamma_0 = 1, \quad \gamma_1 = \frac{\varphi_1}{1 - \varphi_2}, \quad \gamma_2 = \varphi_1\gamma_1 + \varphi_2, \quad \sigma_\epsilon^2 = 1 - \varphi_1\gamma_1 - \varphi_2\gamma_2, \quad (1)$$

all functions of $\varphi = (\varphi_1, \varphi_2)$ alone (`get_ar2_standard_moments`).

- Stationarity restricts φ to the triangle

$$S_2 = \{(\varphi_1, \varphi_2) : \varphi_1 + \varphi_2 < 1, \varphi_2 - \varphi_1 < 1, |\varphi_2| < 1\}. \quad (2)$$

- Under stationarity the joint prior density factorises as

$$p(Y_{1:T} | \varphi) = \underbrace{p(Y_1, Y_2 | \varphi)}_{\text{initial block}} \prod_{t=3}^T N(Y_t; \varphi_1 Y_{t-1} + \varphi_2 Y_{t-2}, \sigma_\epsilon^2), \quad (3)$$

with initial block

$$(Y_1, Y_2)^\top \sim N_2(\mathbf{0}, \Sigma(\varphi)), \quad \Sigma(\varphi) = \begin{pmatrix} 1 & \gamma_1 \\ \gamma_1 & 1 \end{pmatrix}. \quad (4)$$

- Equivalently, $p(Y_1) = N(0, 1)$ and

$$p(Y_2 | Y_1, \varphi) = N(\gamma_1 Y_1, 1 - \gamma_1^2), \quad (5)$$

the “Fix 1” branch of `get_ar2_conditional_params` (see `ar2_fix_proposal.tex`).

2 Exact versus conditional likelihood

Definition 2.1 (Exact and conditional likelihoods). For fixed $Y_{1:T}$, define as functions of $\varphi \in S_2$

$$L_{\text{exact}}(\varphi) = p(Y_{1:T} \mid \varphi), \quad L_{\text{cond}}(\varphi) = p(Y_{3:T} \mid Y_{1:2}, \varphi) = \prod_{t=3}^T N(Y_t; \varphi_1 Y_{t-1} + \varphi_2 Y_{t-2}, \sigma_\epsilon^2). \quad (6)$$

- By (3) the two are linked through the discrepancy factor

$$L_{\text{exact}}(\varphi) = f(\varphi; Y_{1:2}) \cdot L_{\text{cond}}(\varphi), \quad f(\varphi; Y_{1:2}) = N_2((Y_1, Y_2)^\top; \mathbf{0}, \Sigma(\varphi)). \quad (7)$$

- `ar2_transition_loglik` computes $\log L_{\text{cond}}$ only: it sums transition densities over `data[, 3:nt]`, discarding $\log f$ entirely.

3 Why the omission is exact in AR(1)

Lemma 3.1 (AR(1) exemption). *For the unit-variance stationary AR(1), i.e. $p = 1$ with $Y_1 \sim N(0, \gamma_0)$ and $\gamma_0 \equiv 1$, the discrepancy factor satisfies $f(\varphi; Y_1) = N(Y_1; 0, 1)$, free of φ ; hence*

$$L_{\text{exact}}(\varphi) \propto_\varphi L_{\text{cond}}(\varphi) \quad \text{exactly, for every realisation } Y_{1:T}. \quad (8)$$

Proof. The initial block for $p = 1$ is the single marginal $p(Y_1 \mid \varphi) = N(Y_1; 0, \gamma_0)$, and the normalisation $\gamma_0 = 1$ removes its only φ -dependence; (7) with constant f gives (8). $\square$

Remark 3.2 (The original code is correct). The comment in `biometrics/update_params.R` (line 925), “*likelihood of data impacted by phi does not include day 1,*” is therefore an identity under the unit-variance normalisation, not an approximation. `updatePhiTS` and the AR(1) latent blocks consequently share one target posterior, and the original sampler is a valid MH-within-Gibbs scheme.

4 The AR(2) violation

- The sampler is MH-within-Gibbs on (Y, φ) with target

$$\pi(Y, \varphi \mid \text{data}) \propto p(\text{data} \mid Y) p(Y \mid \varphi) p(\varphi), \quad (9)$$

and validity requires every block’s acceptance ratio to be computed under this single π .

- The correct full conditional for φ is, by (3)–(4),

$$\pi(\varphi \mid Y) \propto \underbrace{N_2((Y_1, Y_2)^\top; \mathbf{0}, \Sigma(\varphi))}_{f(\varphi; Y_{1:2})} \prod_{t=3}^T N(Y_t; \varphi_1 Y_{t-1} + \varphi_2 Y_{t-2}, \sigma_\epsilon^2) p(\varphi). \quad (10)$$

Proposition 4.1 (Target mismatch between the two blocks). *Suppose $T \geq 3$ and $p(\varphi) > 0$ on S_2 . In `update_params_ar2.R`:*

- (i) *the Y -block updaters (`updateZTS_AR2`, `updateTauTS_AR2`, `updateKnotsTS_AR2`) evaluate the prior contribution of Y_2 through (5), hence are π -invariant MH kernels;*

(ii) the φ -block (`updatePhiAR2TS`) accepts with ratio built from $L_{\text{cond}}(\varphi) p(\varphi)$, hence is invariant for

$$\tilde{\pi}(Y, \varphi \mid \text{data}) \propto p(\text{data} \mid Y) N(Y_1; 0, 1) L_{\text{cond}}(\varphi) p(\varphi), \quad (11)$$

and $\tilde{\pi} \neq \pi$ because $\pi/\tilde{\pi} \propto f(\varphi; Y_{1:2})/N(Y_2; 0, 1)$ is non-constant in φ whenever $\gamma_1 \neq 0$;

(iii) the composite kernel therefore does not admit π as an invariant distribution.

Proof. Claim (i) is read off the “Fix 1” $t = 2$ branch used in each latent update.

For (ii), $\gamma_1 = \varphi_1/(1 - \varphi_2)$ makes $\Sigma(\varphi)$, hence $f(\varphi; Y_{1:2})$, non-constant on S_2 for any $Y_{1:2}$ with $Y_1 Y_2 \neq 0$ or $Y_1^2 + Y_2^2 \neq 2$: explicitly,

$$\log f(\varphi; Y_{1:2}) = -\log 2\pi - \frac{1}{2} \log(1 - \gamma_1^2) - \frac{Y_1^2 - 2\gamma_1 Y_1 Y_2 + Y_2^2}{2(1 - \gamma_1^2)} \quad (12)$$

depends on φ through γ_1 .

The factor f depends jointly on φ and on $Y_{1:2}$, so it can be absorbed into the normalising constant of neither block’s full conditional.

For (iii), suppose the composite kernel had invariant law π ; since the blocks touch disjoint coordinates, each factor kernel would then be π -invariant on its own coordinates. But the φ -kernel’s unique invariant conditional is $\tilde{\pi}(\varphi \mid Y) \neq \pi(\varphi \mid Y)$, a contradiction. $\square$

Remark 4.2 (Provenance). The port copied the “skip the head of the series” shape of the AR(1) code (Proposition 3.2) without noticing that the skipped term acquires φ -dependence at $p = 2$. Only the Y_1 factor stays exempt: it is $N(0, 1)$ by the $\gamma_0 = 1$ normalisation for every p .

5 Magnitude at the study’s sample size

- The omitted factor (12) carries the information of 2 of the T observations per series, so the perturbation of the φ -posterior is $O(T^{-1})$ with constant governed by $\partial \log f / \partial \varphi$.
- A direct Monte Carlo check ($T = 50$ as in the simulation study, 30 replicate datasets, componentwise MH with the production $N(0, 0.5^2)$ prior and silent rejection on S_2) compares the posterior mean under L_{cond} against L_{exact} ; Table 1 reports the averages.

Table 1: Posterior means of (φ_1, φ_2) under the conditional and exact likelihoods; $T = 50$, 30 replicates, K independent series per dataset. Settings are the fixed-AR(2) DGPs of `simstudy/setup.R`.

Setting	K	Truth	L_{cond} (code)	L_{exact} (with f)
16 near-unit-root	1	(0.15, 0.80)	(0.16, 0.76)	(0.15, 0.77)
16 near-unit-root	5	(0.15, 0.80)	(0.14, 0.81)	(0.13, 0.82)
9 strong	1	(0.80, −0.35)	(0.73, −0.27)	(0.73, −0.27)
9 strong	5	(0.80, −0.35)	(0.78, −0.33)	(0.78, −0.33)
10 weak	1	(0.12, −0.05)	(0.06, −0.10)	(0.06, −0.10)
10 weak	5	(0.12, −0.05)	(0.11, −0.04)	(0.12, −0.04)

Remark 5.1 (Interpretation). All L_{cond} -versus- L_{exact} differences are at most 0.02, within Monte Carlo noise, while the visible departures from truth (0.03–0.07) are the ordinary $O(T^{-1})$ finite-sample AR bias present under both likelihoods. The sampler is thus theoretically invalid but numerically indistinguishable from the corrected one at $T = 50$; the gap scales as $2/T$ and becomes worth re-checking if T drops to 10–20 (time-block forecasting).

6 The fix

- Add the φ -dependent part of $\log f$, namely (5) evaluated at the data, to both `cur.ll` and `can.ll` in `updatePhiAR2TS` (each with its own moments):

```
# inside updatePhiAR2TS, after computing (cur/can).moments:
g1 <- moments$gamma1 # = phi1 / (1 - phi2)
ll <- ll + sum(dnorm(data.t2, g1 * data.t1,
                    sqrt(1 - g1^2), log = TRUE))
# data.t1, data.t2 = the t = 1, 2 slices of 'data'
# (columns for day.mar = 2; third-margin slices for day.mar = 3)
```

- The Y_1 marginal $N(0, 1)$ needs no term: it cancels in the acceptance ratio for every p (Proposition 4.2).
- The silent-rejection proposal scheme is unaffected: Proposition 3.3 of `where_we_are.tex` concerns the proposal's invariance on S_2 and holds for whichever target density is supplied.
- Expected effect: none on the $T = 50$ simulation conclusions (Table 1); the patch restores (10) as the φ -block target and hence validity of the composite sampler.

7 Verification and status

- **Status (2026-07-15): applied.** The patch of Section 6 is in production as `ar2_stationary_loglik` (`auxfunctions_ar2.R`), called at all three likelihood evaluations inside `updatePhiAR2TS` (`update_params_ar2`).
- **Scope of the correctness claim.** The proof of Proposition 4.1 and its resolution never reference the data-generating process, so validity of the fixed sampler holds for every DGP unconditionally; only the *negligibility* claims below are DGP- and T -specific.

7.1 Geweke prior-recovery test (DGP-free)

- Design (successive-conditional simulator, Geweke 2004): switch the data likelihood off, so the target collapses to $p(\varphi)p(Y \mid \varphi)$, and alternate the production Y -block prior structure ($t, t+1, t+2$ companion terms via `ar2_conditional_log_density`) with the production `updatePhiAR2TS`. If and only if the blocks share one joint density must the φ -marginal of the chain equal the $N(0, 0.5^2)^{\otimes 2}$ prior truncated to S_2 (reference: rejection sample, $n \approx 10^6$).
- Small T amplifies the discrepancy: the initial block carries $2/T$ of the information, so $T = 6$ and $T = 4$ were used (1/3 and 1/2 respectively).
- The conditional-only version deviates on $E[\varphi_2]$ in the *same direction* in two independent designs (deviations +0.021 and +0.011; Stouffer-combined $z \approx 2.8$), matching the theory: f acts through $\gamma_1 = \varphi_1/(1 - \varphi_2)$ and so distorts the φ_2 -geometry of the truncated prior first.
- The stationary version passes every statistic in both designs ($|z| \leq 1$), confirming the patch removes the incompatibility in the production function itself, not merely in a reimplementa-tion.

Table 2: Geweke prior-recovery: deviation of the chain’s φ -marginal mean from the truncated-prior reference, in MCMC standard errors (z). “Conditional” = pre-fix production code; “stationary” = patched code.

Design	Version	$z(E[\varphi_1])$	$z(E[\varphi_2])$
$T = 6, K = 3$, one chain of 115k	conditional	1.0	2.1
	stationary	0.4	−0.7
$T = 4, K = 3$, six chains of 40k	conditional	0.5	1.8
	stationary	0.1	−0.6

7.2 DGP coverage of the negligibility claim

- Table 1 covers the fixed-AR(2) DGPs (settings 9–16); Table 3 extends the same comparison ($T = 50, K = 1, 30$ replicates) to the remaining DGP families of `simstudy/setup.R`.
- For the long-memory rows the reference is the Yule–Walker pseudo-true projection `arfima_yw_projection(d)` the residual gap to it is finite-sample bias, identical under both likelihoods.

Table 3: Conditional-versus-exact posterior means on the remaining DGP families; last column is the largest coordinatewise difference.

DGP	Pseudo-true	L_{cond}	L_{exact}	max diff
White noise (settings 1–8)	(0.00, 0.00)	(0.02, 0.01)	(0.02, 0.01)	0.004
ARFIMA $d = 0.20$ (17)	(0.22, 0.11)	(0.18, 0.08)	(0.18, 0.08)	0.001
ARFIMA $d = 0.35$ (18)	(0.42, 0.21)	(0.34, 0.19)	(0.34, 0.19)	0.001
ARFIMA $d = 0.45$ (19)	(0.58, 0.29)	(0.59, 0.25)	(0.58, 0.25)	0.003

7.3 Permanent regression guard

- `tests/test_phi_block_consistency.R` now protects the property in three layers: a structural assertion (`updatePhiAR2TS` calls `ar2_stationary_loglik` and never `ar2_transition_loglik`), an exact identity check (the added term equals the latent-block $t = 2$ density (5) to 10^{-10} on 200 random draws), and a fixed-seed Geweke smoke gate ($|z| < 4$).
- Layer two is the theorem-level guard — it verifies block compatibility directly and costs milliseconds; the heavier detection of Table 2 is documented here rather than re-run per test invocation.